

Class Assignment 3

Business Intelligence

Topics: Principal Component Analysis + Decision Trees

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Part I. Principal Component Analysis (PCA)

Business Context: E-Commerce Customer Engagement Strategy

Vanguard Retail Group has tracked $n = 15,000$ high-frequency customers to optimize their loyalty program. The data science team identified five key performance indicators (KPIs) related to spending and digital interaction:

- $\mathbf{X}_1$: Annual Spending (USD).
- $\mathbf{X}_2$: Average Session Duration (minutes).
- $\mathbf{X}_3$: Historical Purchase Count (units).
- $\mathbf{X}_4$: Technical Support Interactions (tickets).
- $\mathbf{X}_5$: Promotion Response Rate (percentage).

All variables were standardized ($\mu = 0, \sigma = 1$) to handle different scales. A Principal Component Analysis (PCA) was performed to simplify these metrics into strategic indices.

Correlation Matrix ($\mathbf{R}$)

The following matrix shows significant positive correlations ($r > 0.70$) between spending metrics (X_1, X_3, X_5) and between digital engagement metrics (X_2, X_4), justifying the use of PCA to reduce redundancy.

$$\mathbf{R} = \begin{pmatrix} 1.00 & 0.22 & 0.78 & 0.15 & 0.82 \\ 0.22 & 1.00 & 0.12 & 0.88 & 0.18 \\ 0.78 & 0.12 & 1.00 & 0.08 & 0.71 \\ 0.15 & 0.88 & 0.08 & 1.00 & 0.10 \\ 0.82 & 0.18 & 0.71 & 0.10 & 1.00 \end{pmatrix}$$

Eigenvalues and Variance Explained

Based on the total system variance ($n = 5$ variables), the following components were extracted.

Component	Eigenvalue (λ_i)	Proportion of Var.	Cumulative %
Z_1 (PC ₁)	2.720	0.5440	54.40%
Z_2 (PC ₂)	1.680	0.3360	88.00%
Z_3 (PC ₃)	0.320	0.0640	94.40%
Z_4 (PC ₄)	0.190	0.0380	98.20%
Z_5 (PC ₅)	0.090	0.0180	100.00%

Full Loading Matrix (Component Matrix)

This matrix represents the correlation between the original KPIs ($\mathbf{X}_n$) and all extracted principal components (Z_1 through Z_5).

Variable	$Z_1(PC_1)$	$Z_2(PC_2)$	$Z_3(PC_3)$	$Z_4(PC_4)$	$Z_5(PC_5)$
$\mathbf{X}_1$: Annual Spending	0.912	-0.105	0.224	-0.118	0.052
$\mathbf{X}_2$: Session Duration	0.184	0.892	-0.315	0.142	-0.221
$\mathbf{X}_3$: Purchase Count	0.845	-0.098	-0.422	0.310	0.084
$\mathbf{X}_4$: Support Interactions	0.122	0.876	0.284	-0.215	0.298
$\mathbf{X}_5$: Promotion Response	0.889	0.145	0.102	0.126	-0.395

Multiple-Choice Questions (2-points each)

Answer each question by selecting the **single best** option.

1. When PCA is performed on the correlation matrix $\mathbf{R}$ of the 5 marketing KPIs, the sum of all eigenvalues ($\sum \lambda_i$) equals:

- a) 5.0 (The number of variables).
- b) 1.0 (The standard variance).
- c) 15,000 (The number of customers).
- d) 0.88 (The cumulative variance).

2. According to the **Kaiser Criterion**, how many principal components should be retained for this marketing analysis?

- a) 1
- b) 2
- c) 3
- d) 5

3. The principal components Z_i and Z_j (for $i \neq j$) have a correlation of zero. This algebraic property is known as:

- a) Collinearity
- b) Orthogonality
- c) Heteroscedasticity
- d) Linearity

4. Which set of variables loads most strongly (highest absolute correlation) on the first component Z_1 ?

- a) X_2 and X_4
- b) X_1 , X_3 , and X_5
- c) X_1 and X_2
- d) X_3 and X_4

5. In the context of the Machine Learning taxonomy, into which category does the Principal Component Analysis (PCA) algorithm fall?

- a) Supervised Learning — Regression
- b) Supervised Learning — Classification
- c) Unsupervised Learning — Dimensionality Reduction
- d) Unsupervised Learning — Clustering

6. The cumulative proportion of variance explained by the first two components (Z_1, Z_2) is 88.00%. This means:

- a) 88.00% of the customers respond to promotions.
- b) 88.00% of the total information (variability) of the 5 KPIs is captured by these two indices.
- c) 88.00% of the correlations in $\mathbf{R}$ are positive.
- d) The model has an error rate of 12.00%.

7. Based on its loadings, Z_2 can be best interpreted as an index of:

- a) Financial capacity and transactional volume.
- b) Digital engagement and platform interaction.
- c) Marketing cost and ROI.
- d) Customer tenure and loyalty.

8. What is the primary objective of applying PCA to Vanguard's customer data?

- a) To identify which customers are likely to churn.
- b) To reduce the complexity of 5 KPIs into 2 strategic dimensions while minimizing information loss.
- c) To calculate the average annual spending of the 15,000 customers.
- d) To group customers into segments with similar behaviors.

9. Vanguard's data science team chose PCA over a simple variable selection (Feature Selection). What is the primary advantage of this approach?

- a) It preserves the original physical meaning of each KPI for easier reporting.
- b) It creates new synthetic features that capture the maximum possible information from all original KPIs.
- c) It eliminates the need to standardize variables with different scales.
- d) It guarantees that the resulting components will be highly correlated with each other.

10. In the context of Vanguard's dataset, why must variables like "Annual Spending" (measured in USD) and "Promotion Response Rate" (measured in percentage) be standardized before running PCA?

- a) To ensure the sample size n reaches 15,000 observations.
- b) To prevent variables with larger units or scales (like USD) from dominating the model variance and the principal components.
- c) To convert quantitative spending data into qualitative categorical labels.
- d) To make the mean of each variable equal to the total number of components extracted.

Part II. Decision Trees

A major bank processes **10,000 credit card transactions per hour**. The fraud prevention team has built a classification decision tree to flag potentially **fraudulent transactions** in real time. Catching fraud early saves the bank an average of \$1,200 per fraudulent transaction; however, incorrectly blocking a legitimate transaction costs \$15 in customer service and damages trust. The model was trained on **600 historical transactions**.

Variable	Description	Type
transaction_amount	Amount of the transaction (USD)	Continuous
distance_from_home	Distance from cardholder's home (km)	Continuous
is_fraud	1 = Fraudulent, 0 = Legitimate	Binary (target)

In the training data, **420 transactions were legitimate (L)** and **180 were fraudulent (F)**.

Reference Formulas

Gini Impurity:

$$Gini(t) = 1 - \sum_{k=1}^K p_k^2$$

Weighted Gini after a split:

$$Gini_{\text{split}} = \frac{N_L}{N} Gini(\text{left}) + \frac{N_R}{N} Gini(\text{right})$$

Information Gain:

$$\Delta Gini = Gini(\text{parent}) - Gini_{\text{split}}$$

Leaf prediction: majority class

Gini range (binary): [0, 0.5]

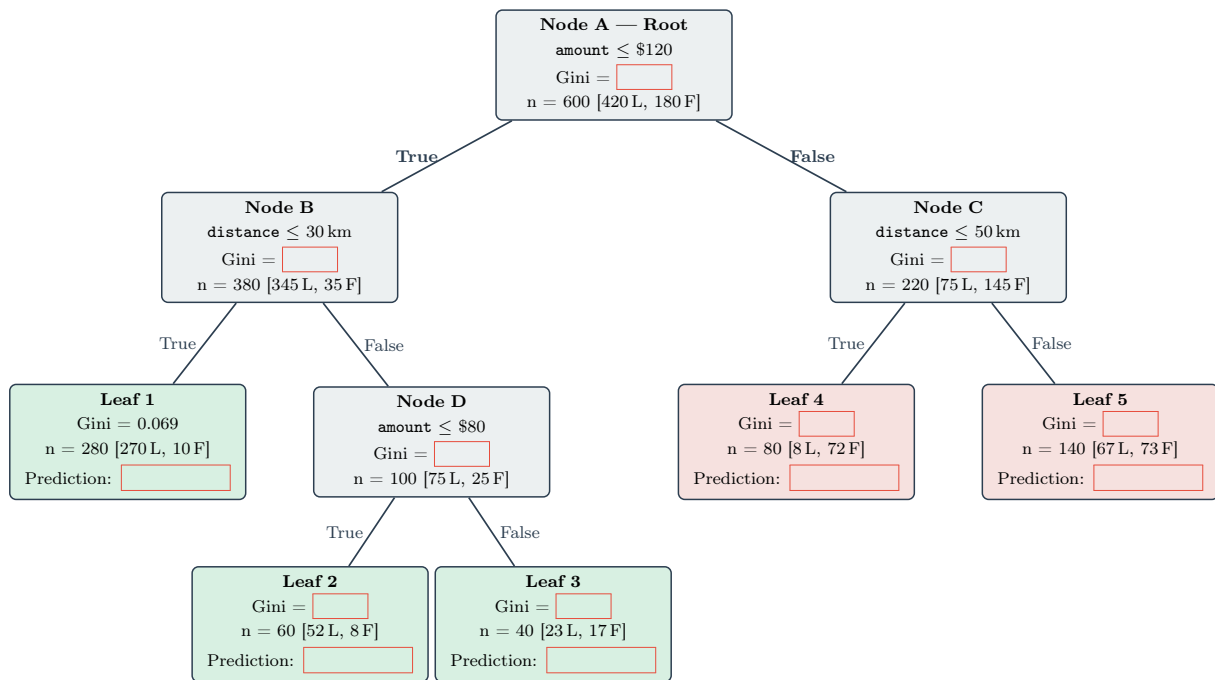


Figure 1: Classification tree for fraud detection. Red boxes = values to be calculated.

Question 1 (9 pts) Complete the missing Gini values and predictions in Figure 1. Fill in the summary table below.

Node/Leaf	Legit	Fraud	Gini Index	Predicted Class
Node A (Root)	420	180		
Node B	345	35		
Node C	75	145		
Leaf 1	270	10		
Node D	75	25		
Leaf 2	52	8		
Leaf 3	23	17		
Leaf 4	8	72		
Leaf 5	67	73		

Question 2 (4 pts) Calculate the information gain ($\Delta Gini$) of the root split (Node A $\rightarrow$ Node B + Node C). Is this a good first split? Justify your answer by interpreting the magnitude of the gain.

Question 4 (4 pts) A new transaction arrives: `amount` = \$200 and `distance` = 40 km. Trace the path in the tree and state the final prediction. Based on the Gini index of the leaf where it lands, how confident should the bank be in this specific decision?.

Question 4 (4 pts) The team compares this tree (depth=4) with a much deeper one (depth=15). The deeper tree has 99% accuracy on training data but only 72% on test data. Explain which tree you would deploy and why, using the concept of **overfitting**.

Question 5 (4 pts) Focus on **Leaf 5**. Using the business costs provided (\$1,200 saved per caught fraud, \$15 cost per false alarm), calculate the net financial benefit for the 140 transactions in this leaf if the bank blocks all of them. Is it a good business decision?.